

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-2: Digital Image Fundamentals

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-1&2)

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6. Representing Digital Images

Assume that an image $f(x, y)$ is sampled so that the resulting digital image has M rows and N columns.

The values of the coordinates (x, y) now become discrete quantities. Thus, the values of the coordinates at the origin are $(x, y)=(0, 0)$.

The next coordinate values along the first row of the image are represented as $(x, y)=(0, 1)$.

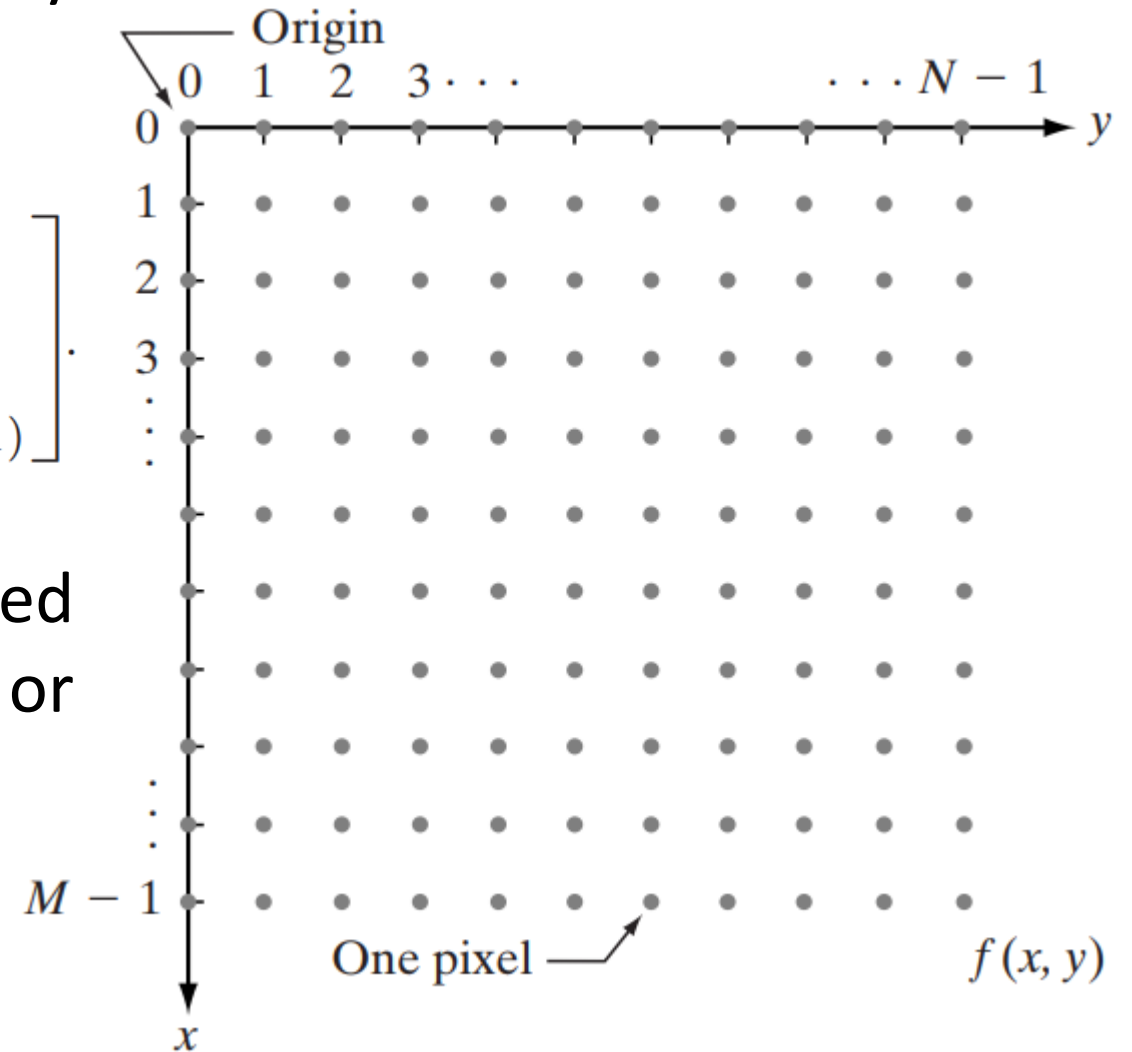
It is important to keep in mind that the notation $(0, 1)$ is used to signify the second sample along the first row. It does not mean that these are the actual values of physical coordinates when the image was sampled.

The right side of this equation is by definition a digital image.

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \cdots & f(0, N - 1) \\ f(1, 0) & f(1, 1) & \cdots & f(1, N - 1) \\ \vdots & \vdots & & \vdots \\ f(M - 1, 0) & f(M - 1, 1) & \cdots & f(M - 1, N - 1) \end{bmatrix}.$$

Each element of this matrix array is called an image element, picture element, pixel, or pel.

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}.$$



This digitization process requires decisions about values for M, N, and for the number, L, of discrete gray levels allowed for each pixel.

There are no requirements on M and N, other than that they have to be positive integers.

However, due to processing, storage, and sampling hardware considerations, the number of gray levels typically is an integer power of 2:

$$L = 2^k \quad \text{k-bit image}$$

k is the number of bits used to represent each pixel in the image.

1-bit Image ($k = 1$):

$$L = 2^1 = 2$$

The image can represent only 2 gray levels: black (0) and white (1).

2-bit Image ($k = 2$):

$$L = 2^2 = 4$$

The image can represent 4 gray levels: 0 (black), 1 (dark gray), 2 (light gray), and 3 (white).

3-bit Image ($k = 3$):

$$L = 2^3 = 8$$

The image can represent 8 gray levels ranging from 0 (black) to 7 (white), with intermediate shades of gray.

8-bit Image ($k = 8$):

$$L = 2^8 = 256$$

The image can represent 256 gray levels, ranging from 0 (black) to 255 (white). This is a common standard for grayscale images.

We assume that the discrete levels are equally spaced and that they are integers in the interval $[0, L-1]$.

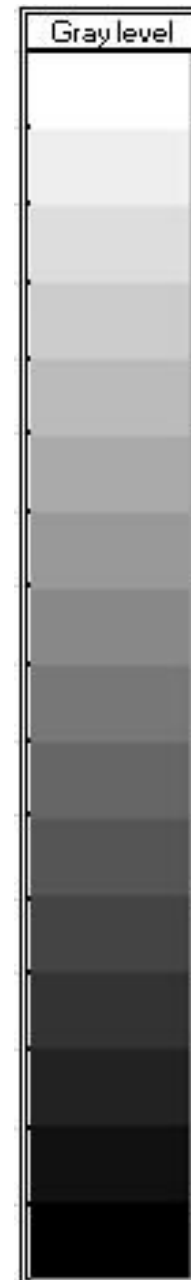
Sometimes the range of values spanned by the gray scale is called the dynamic range of an image

The number of bits (b) required to store a digitized image is

$$b = M \times N \times k.$$

When $M=N$, this equation becomes

$$b = N^2 k.$$



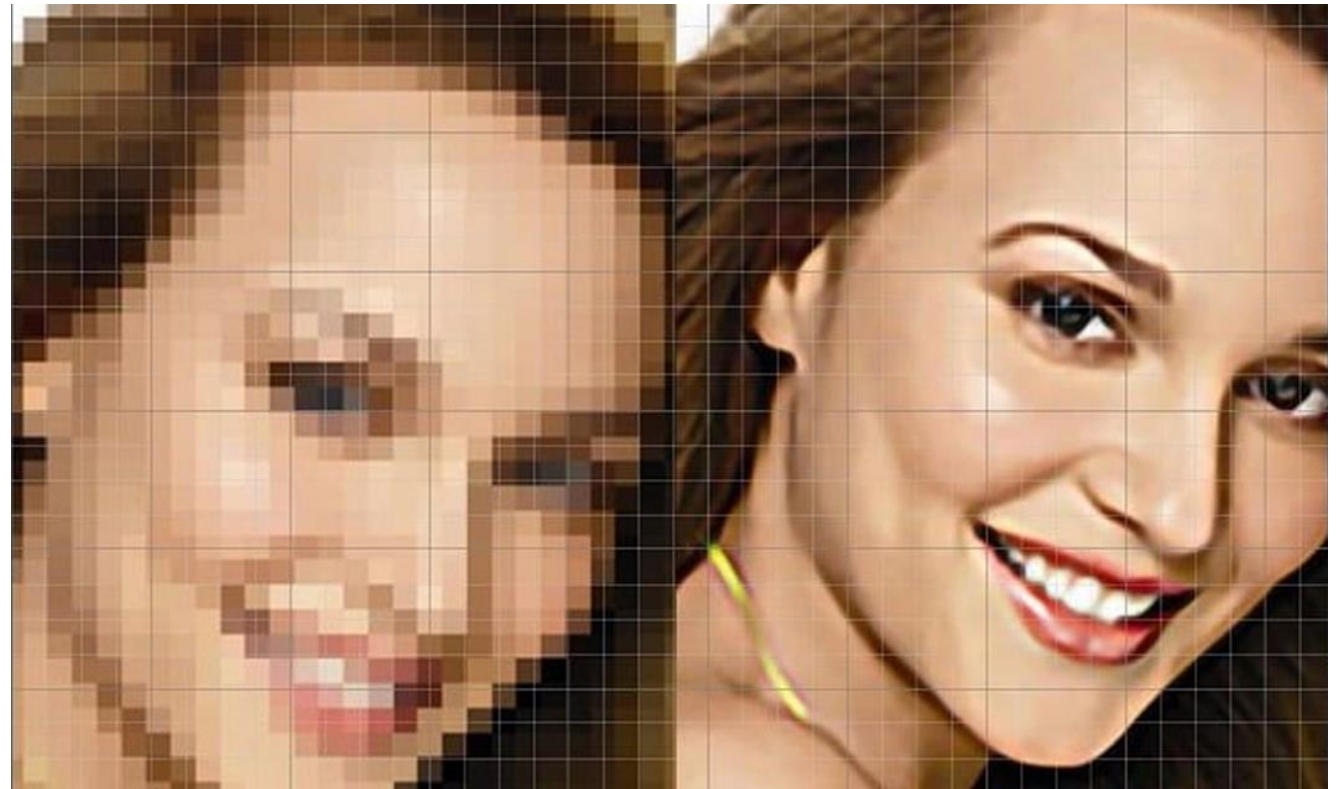
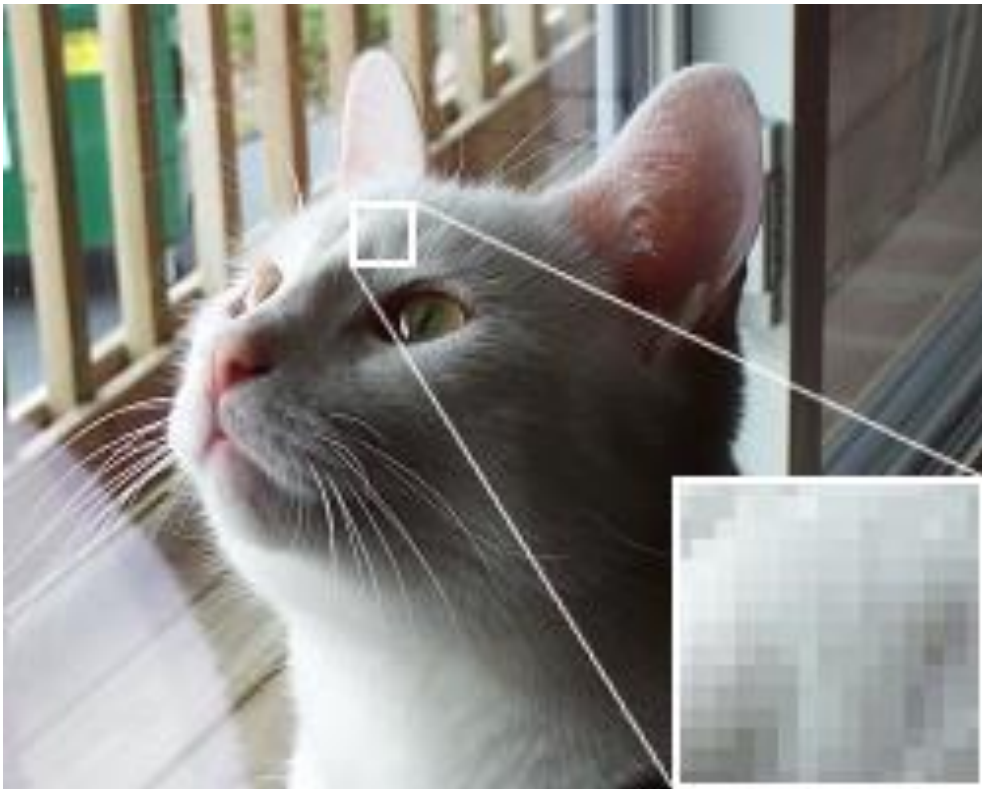
Gray level	Dec	Hex	Bin
	0	00	0000
	1	01	0001
	2	02	0010
	3	03	0011
	4	04	0100
	5	05	0101
	6	06	0110
	7	07	0111
	8	08	1000
	9	09	1001
	10	0A	1010
	11	0B	1011
	12	0C	1100
	13	0D	1101
	14	0E	1110
	15	0F	1111

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

6.1 Pixelation

Pixelation refers to the visual effect that occurs when an image is displayed or zoomed in so much that individual pixels (the smallest units of the image) become visible.



6.2 Spatial Resolution

It defines how well an imaging system can distinguish small objects or details that are close together.

At low resolution, zooming makes the image pixelated, with blocks of colors instead of details.



low spatial resolution



medium spatial resolution



high spatial resolution

At high resolution, zooming still shows sharp details without pixelation.



High Spatial Resolution



Medium Spatial Resolution

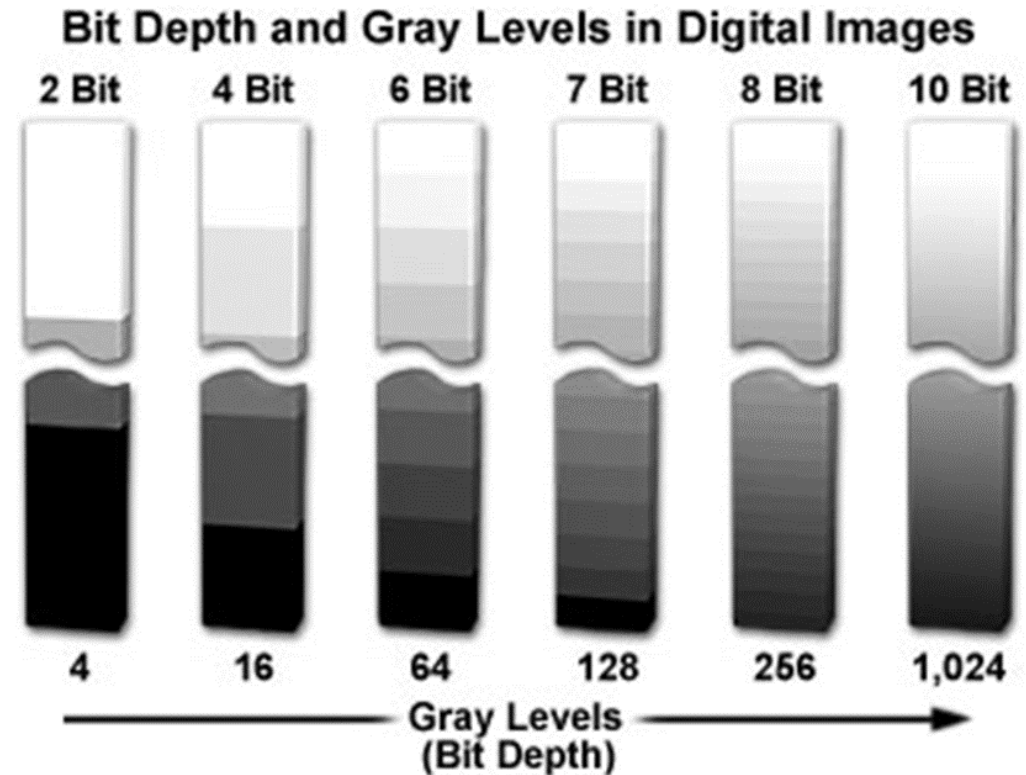


Low Spatial Resolution

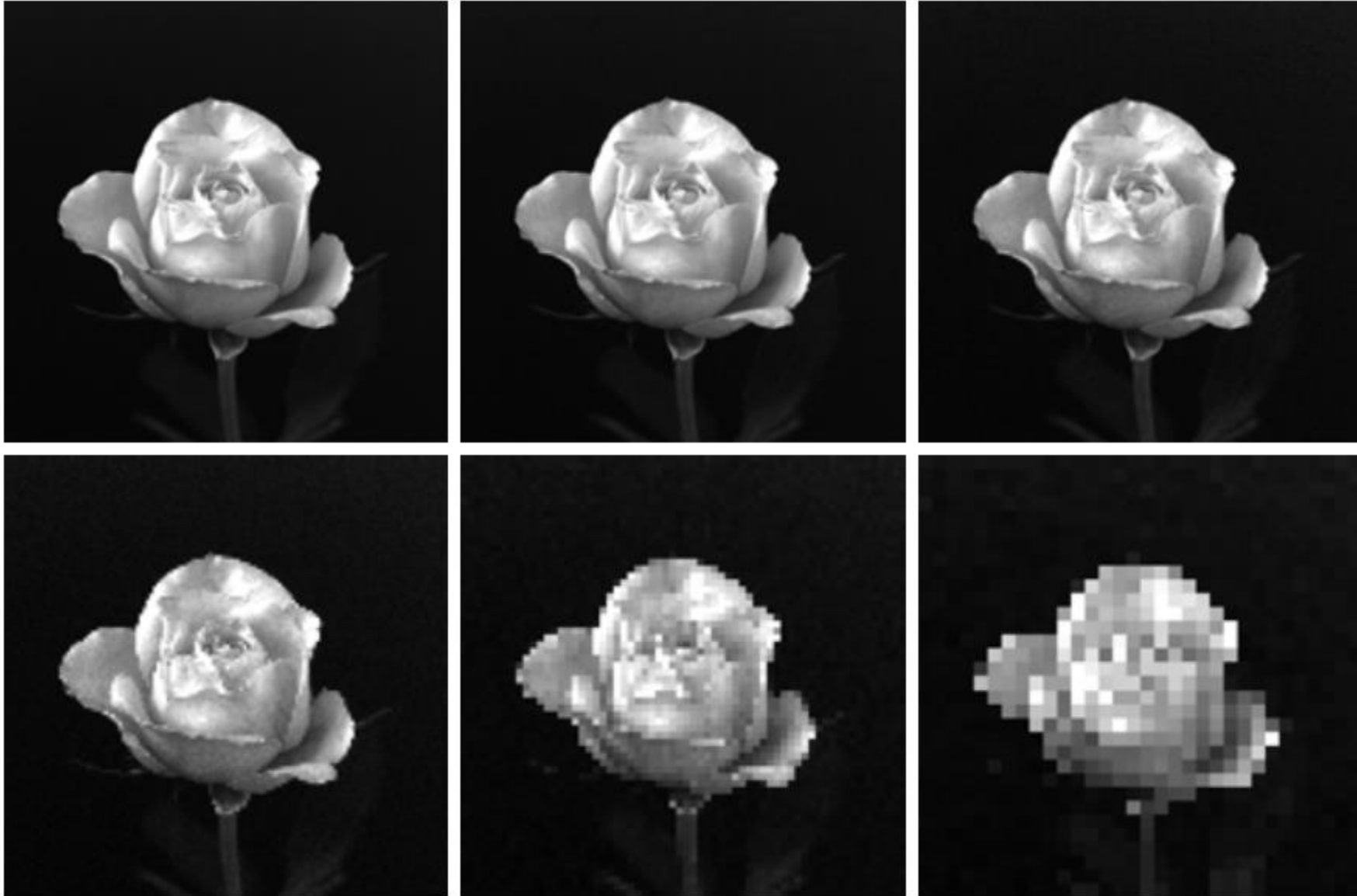
6.3 Gray-level resolution

Gray-level resolution refers to the smallest distinguishable difference in brightness or intensity between two pixels in a digital image.

k=8 (256 gray levels)



1024*1024



32*32

6.4 Isopreference curves

For images with a large amount of detail only a few gray levels may be needed.



Image with a low level of detail



Image with a medium level of detail



Image with a relatively large amount of detail.

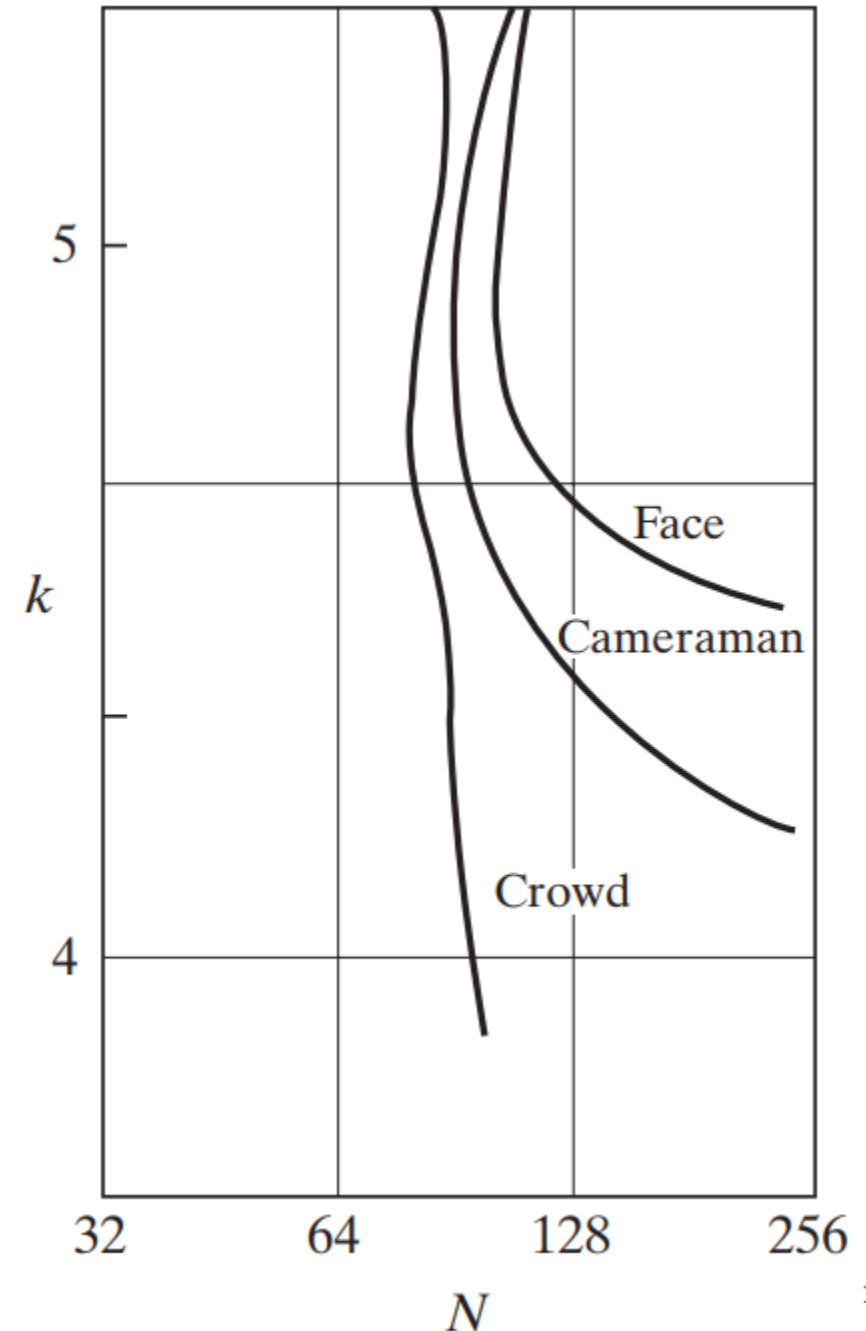
For example, the isopreference curve corresponding to the crowd is nearly vertical.

This indicates that, for a fixed value of N , the perceived quality for this type of image is nearly independent of the number of gray levels used

$$b = M \times N \times k.$$

When $M = N$, this equation becomes

$$b = N^2 k.$$



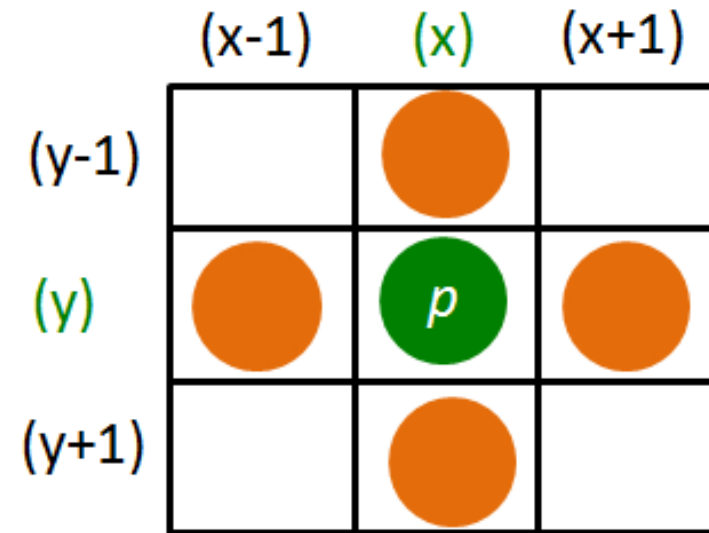
7. Relationships Between Pixels

7.1 Neighbors of a Pixel

A pixel p at coordinates (x, y) has four horizontal and vertical neighbors whose coordinates are given by

$$(x + 1, y), (x - 1, y), (x, y + 1), (x, y - 1)$$

This set of pixels, called the 4-neighbors of p , is denoted by $N_4(p)$.



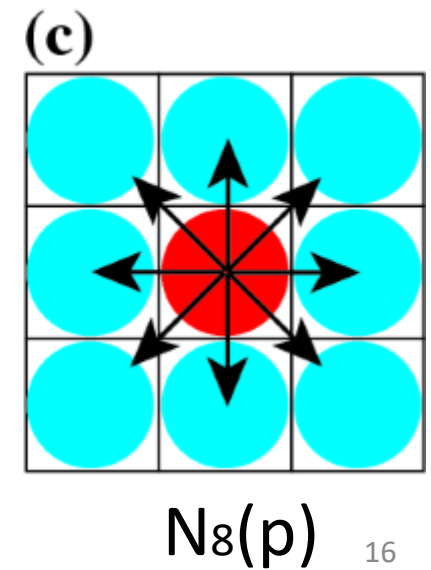
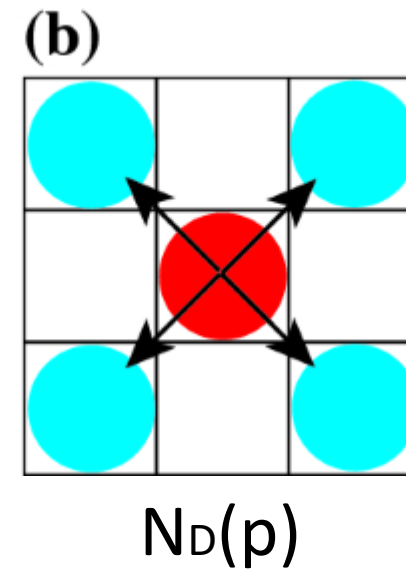
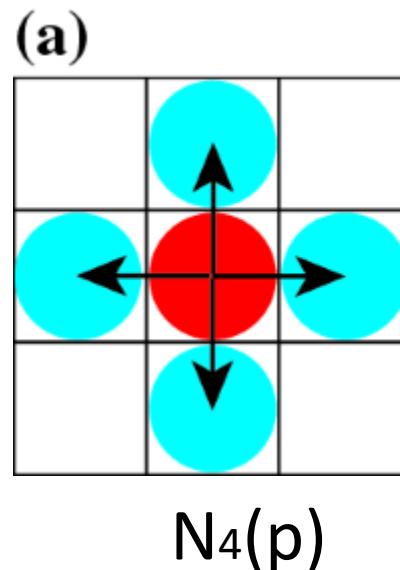
4-neighbors $N_4(p)$

The four diagonal neighbors of p have coordinates

$$(x + 1, y + 1), (x + 1, y - 1), (x - 1, y + 1), (x - 1, y - 1)$$

and are denoted by $N_D(p)$.

These points, together with the 4-neighbors, are called the 8-neighbors of p , denoted by $N_8(p)$.



7.2 Adjacency

Adjacency defines how pixels relate to their neighboring pixels.

Two pixels are considered adjacent if they share a common edge or corner, depending on the type of adjacency used.

Let V be the set of gray-level values used to define adjacency.

- ❑ **In a binary image**, $V=\{1\}$ if we are referring to adjacency of pixels with value 1.
- ❑ **In a grayscale image**, the idea is the same, but set V typically contains more elements. For example, in the adjacency of pixels with a range of possible gray-level values 0 to 255, set V could be any subset of these 256 values.

7.2.1 Types of Adjacency

4-adjacency:

Two pixels p and q with values from V are 4-adjacent *if q is in the set $N_4(p)$.*

For a pixel p , the adjacent pixels are $(x-1, y)$, $(x+1, y)$, $(x, y-1)$, $(x, y+1)$.

A pixel is *adjacent to its immediate left, right, top, and bottom neighbors.*

	4	
4	P	4
	4	

8-adjacency:

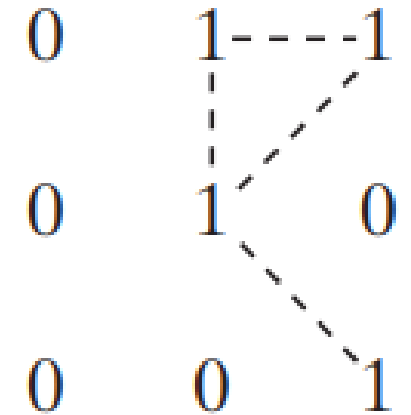
Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

A pixel is adjacent to all its immediate neighbors, including diagonals.

For p , the adjacent pixels include $(x-1,y)$, $(x+1,y)$, $(x,y-1)$, $(x,y+1)$ and $(x-1,y-1)$, $(x,y-1)$, $(x+1,y-1)$.

D	4	D
4	P	4
D	4	D

8	8	8
8	P	8
8	8	8



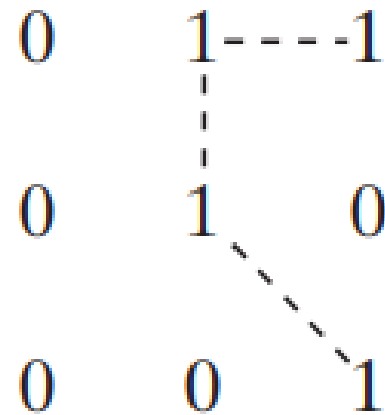
$$8 = 4 \cup D$$

m-adjacency (mixed adjacency):

Two pixels p and q with values from V are m-adjacent if

(i) q is in $N_4(p)$, **or**

(ii) q is in $N_D(p)$ **and** the set has $N_4(p) \cap N_4(q)$ no pixels whose values are from V .



Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.

For example, consider the pixel arrangement shown in (a) for $V = \{1\}$. The three pixels at the top of Fig. (b) show multiple (ambiguous) 8-adjacency, as indicated by the dashed lines.

0	1	1
0	1	0
0	0	1

a b c

0	1	1
0	1	0
0	0	1

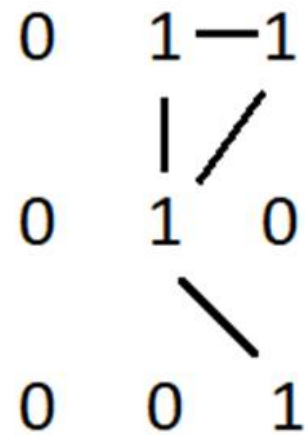
0	1	1
0	1	0
0	0	1

This ambiguity is removed by using m-adjacency, as shown in Fig. (c)

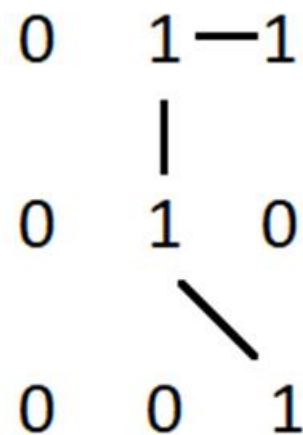
7.3 Connectivity

To establish if two pixels are connected, it must be determined if they are neighbors and if their gray levels satisfy a specified criterion of similarity (say, if their gray levels are equal).

For instance, in a binary image with values 0 and 1, two pixels may be 4-neighbors, but they are said to be connected only if they have the same value.



8-connectivity



m-connectivity

7.3.1 Types of Connectivity

4-connectivity:

Pixels connected through 4-adjacency. Useful for defining regions where diagonal connections are not allowed.

8-connectivity:

Pixels connected through 8-adjacency. This includes diagonal connections, allowing broader region formation.

m-connectivity:

Ensures connectivity without diagonal ambiguities by combining 4-connectivity and 8-connectivity.


```

[ 0 0 0 0 0 0 0 0 ;
  0 1 1 0 0 1 1 1 ;
  0 1 1 0 0 0 1 1 ;
  0 1 1 0 0 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 0 0 0 0 0 ];

```

4-Connected Components

```

[ 0 0 0 0 0 0 0 0 ;
  0 1 1 0 0 1 1 1 ;
  0 1 1 0 0 0 1 1 ;
  0 1 1 0 0 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 1 1 0 0 0 ;
  0 0 0 0 0 0 0 0 ];

```

8-Connected Components

7.4 Regions

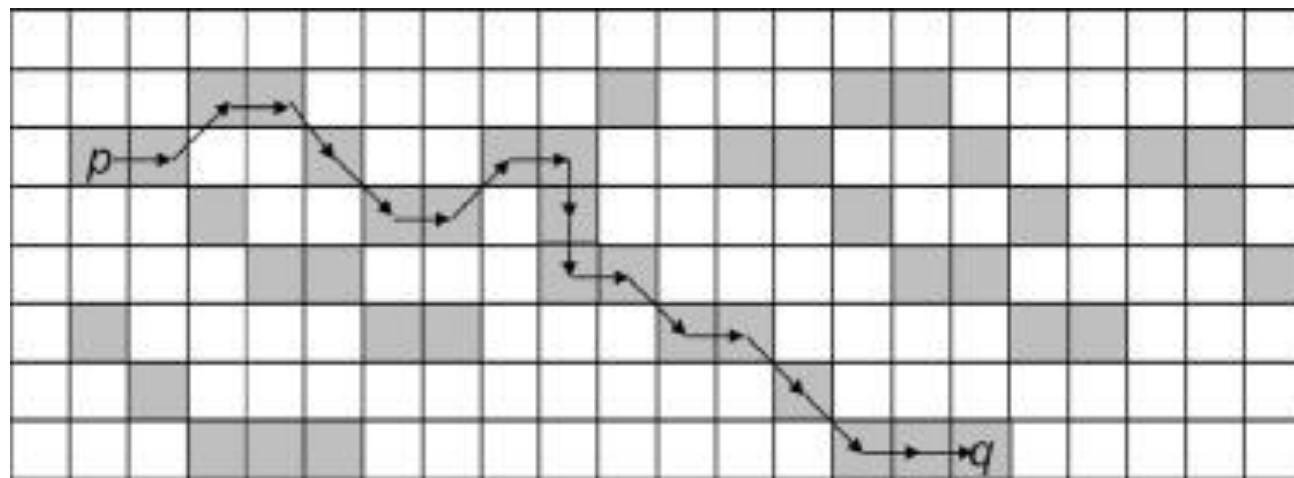
Regions in an image are defined as connected groups of pixels that share a common property, such as intensity or color.

Let R be a subset of pixels in an image. We call R a region of the image if R is a connected set.

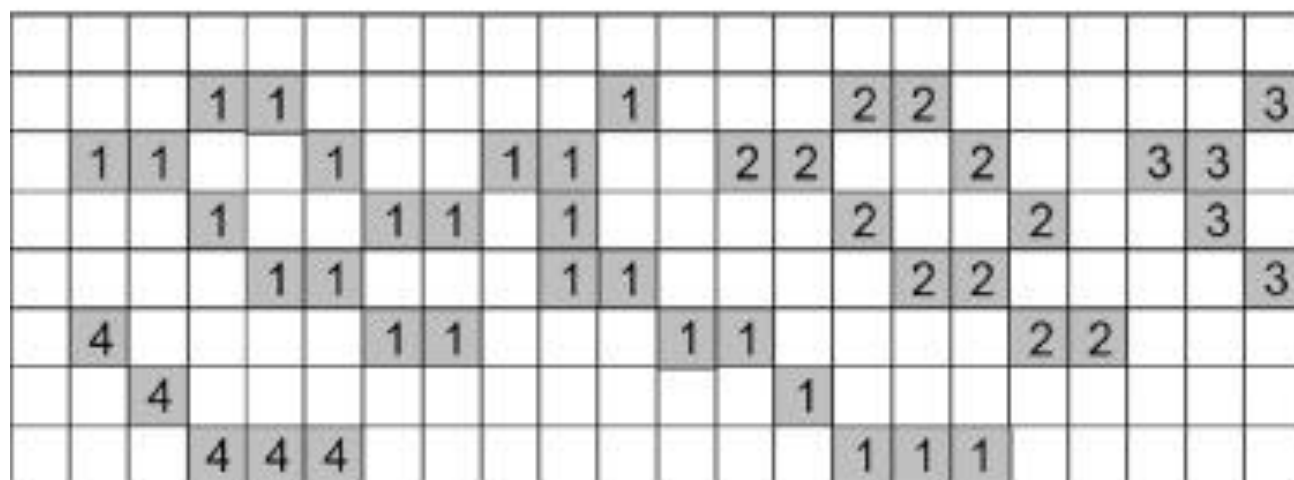
Properties: A region is a set of connected pixels. The background is the complement of a region.

Homogeneity: Pixels in a region have similar properties.

In a binary image, all pixels with value 1 that are connected form a region.



(a)



(b)



background pixel



object pixel

7.5 Boundaries

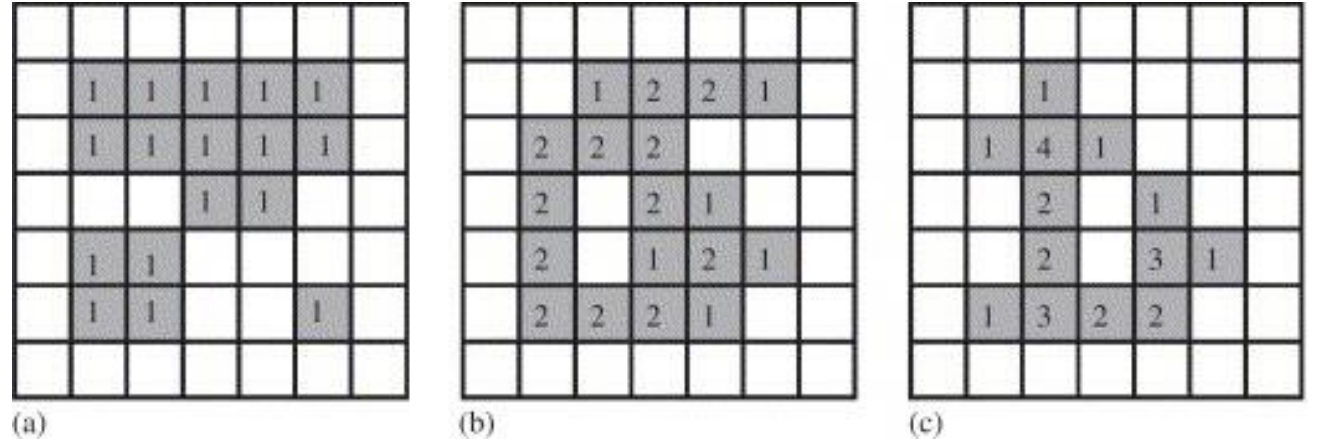
The boundary (also called border or contour) of a region R is the set of pixels in the region $that have one or more neighbors$ that are not in R .

Boundaries (or edges) are the lines or curves that separate different regions in an image.

7.5.1 Types of Boundaries:

Internal Boundary:

Lies entirely within a region, separating pixels of the region from the background or neighboring regions.



External Boundary:

Surrounds the region, defining the interface between the region and its surroundings.

Example: In a segmented image of objects, the boundary of an object separates it from the background.